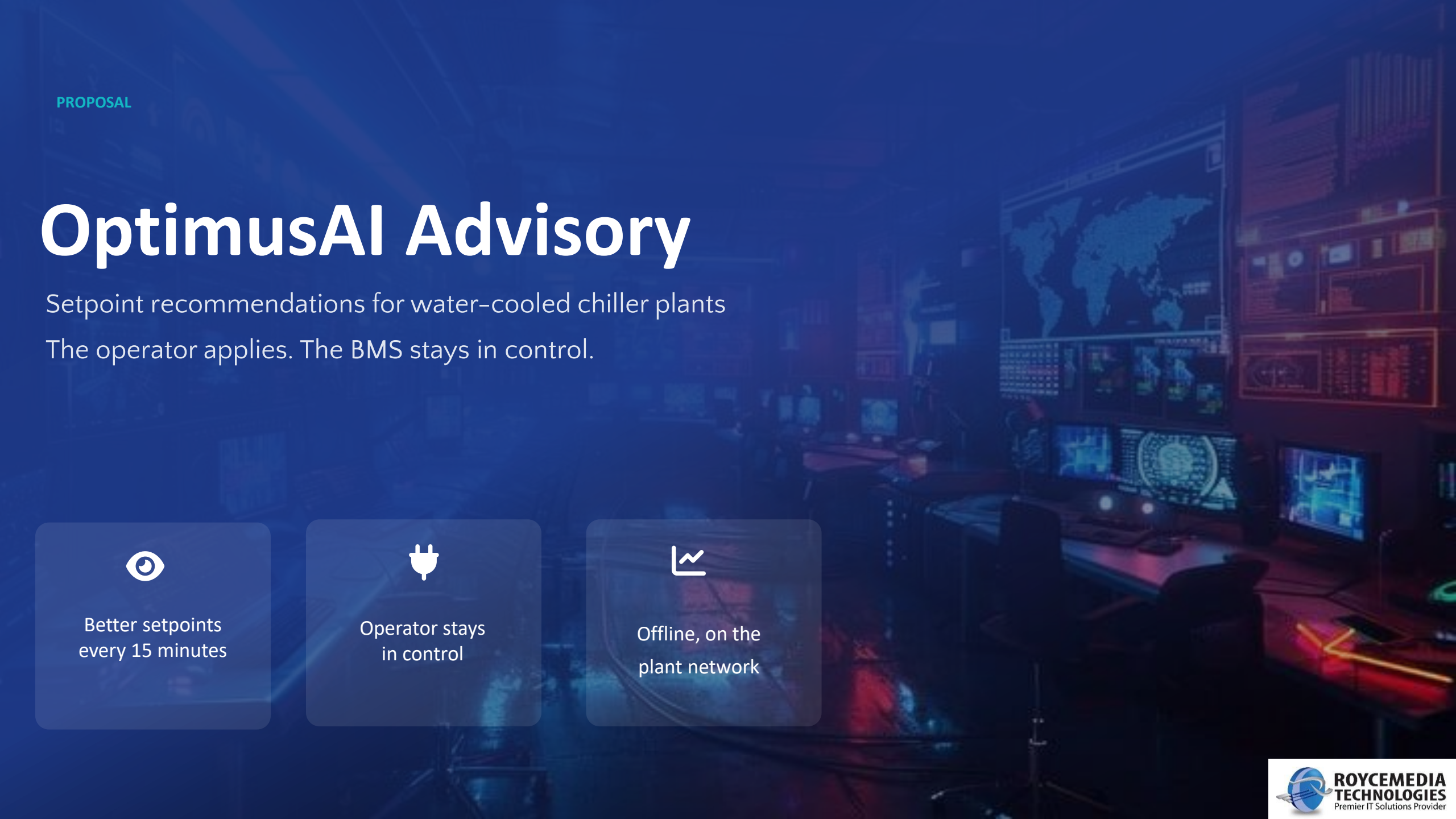
PROPOSAL

OptimusAI Advisory

Setpoint recommendations for water-cooled chiller plants

The operator applies. The BMS stays in control.



Better setpoints every 15 minutes

Operator stays in control

Offline, on the plant network

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Fixed setpoints all day, whatever the plant needs

Chilled water held at 10.0 °C

- For most of the day the cooling coils do not need water that cold: their valves are only partly open.
- Each degree warmer cuts chiller power by about 2 to 3 %.

Tower setpoint stuck at 29.5 °C

- The BMS formula 0.5 × wet-bulb + 27 lands above its own 29.5 °C maximum for Singapore wet-bulbs, so the towers never run colder.
- Each degree colder cuts chiller power by about 3 %, for a little more fan power.

What the advisory system recommends instead

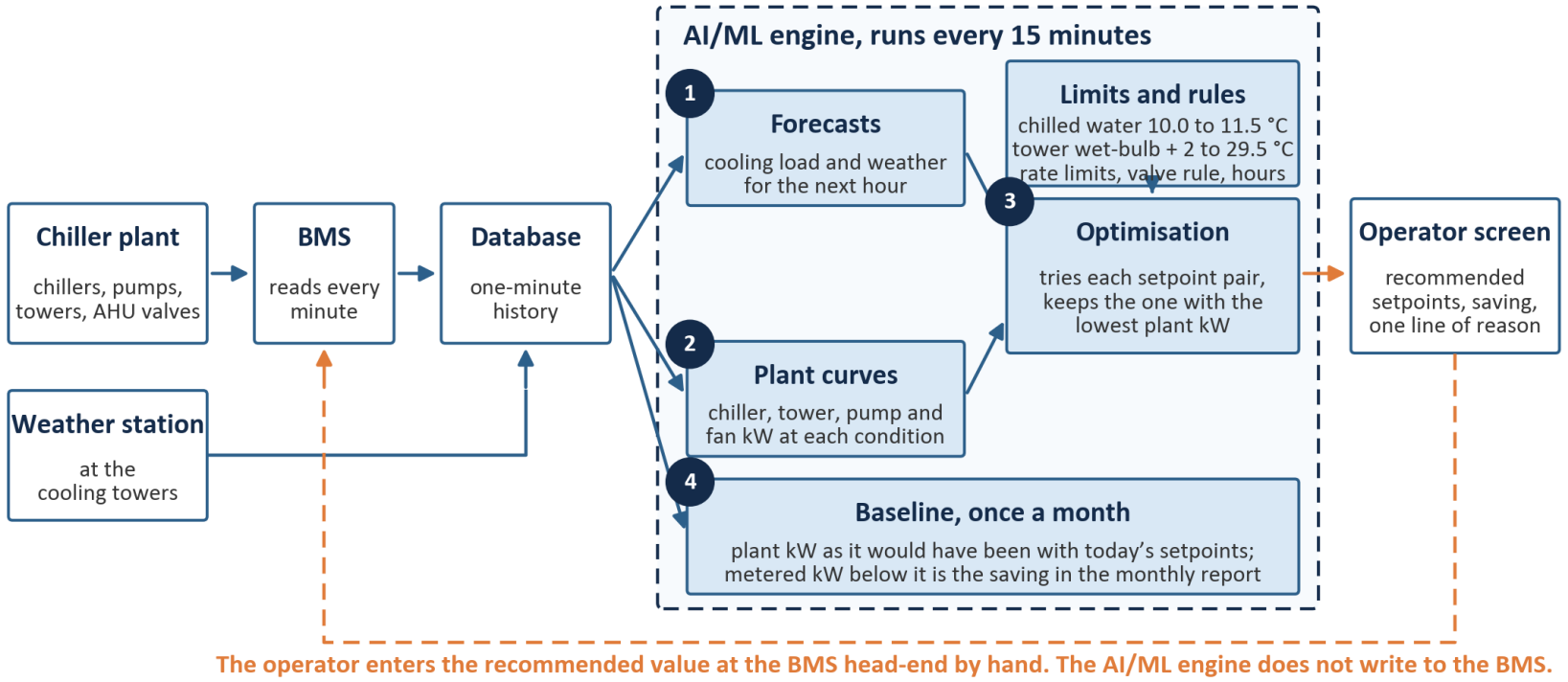
Value	Today	Recommended range	Expected effect
Chilled water supply setpoint	Fixed at 10.0 °C	10.0 to 11.5 °C, up to 0.3 °C every 15 min	Chiller power down 2 to 3 % per degree raised; pumps up a little
Cooling tower leaving water setpoint	29.5 °C in practice	Wet-bulb + 2 °C up to 29.5 °C, up to 0.5 °C every 15 min	Chiller power down about 3 % per degree lowered; fans up

The rule in one line. Every 15 minutes, find the chilled water and tower setpoints that give the lowest plant kW while every cooling coil stays satisfied and the towers can still reach the water temperature asked of them.

Two to eight recommendations a day. Each is one number for the operator to key in at the BMS head-end.

Percentages are from published studies of plants like this; the plant's own figures come from its data once the models are fitted.

How it works: read, forecast, price each setpoint, recommend



1 Read

Every point on the BMS, once a minute. Nothing is written back.

2 Forecast

Cooling load and weather for the next hour, from the plant's own history.

3 Price

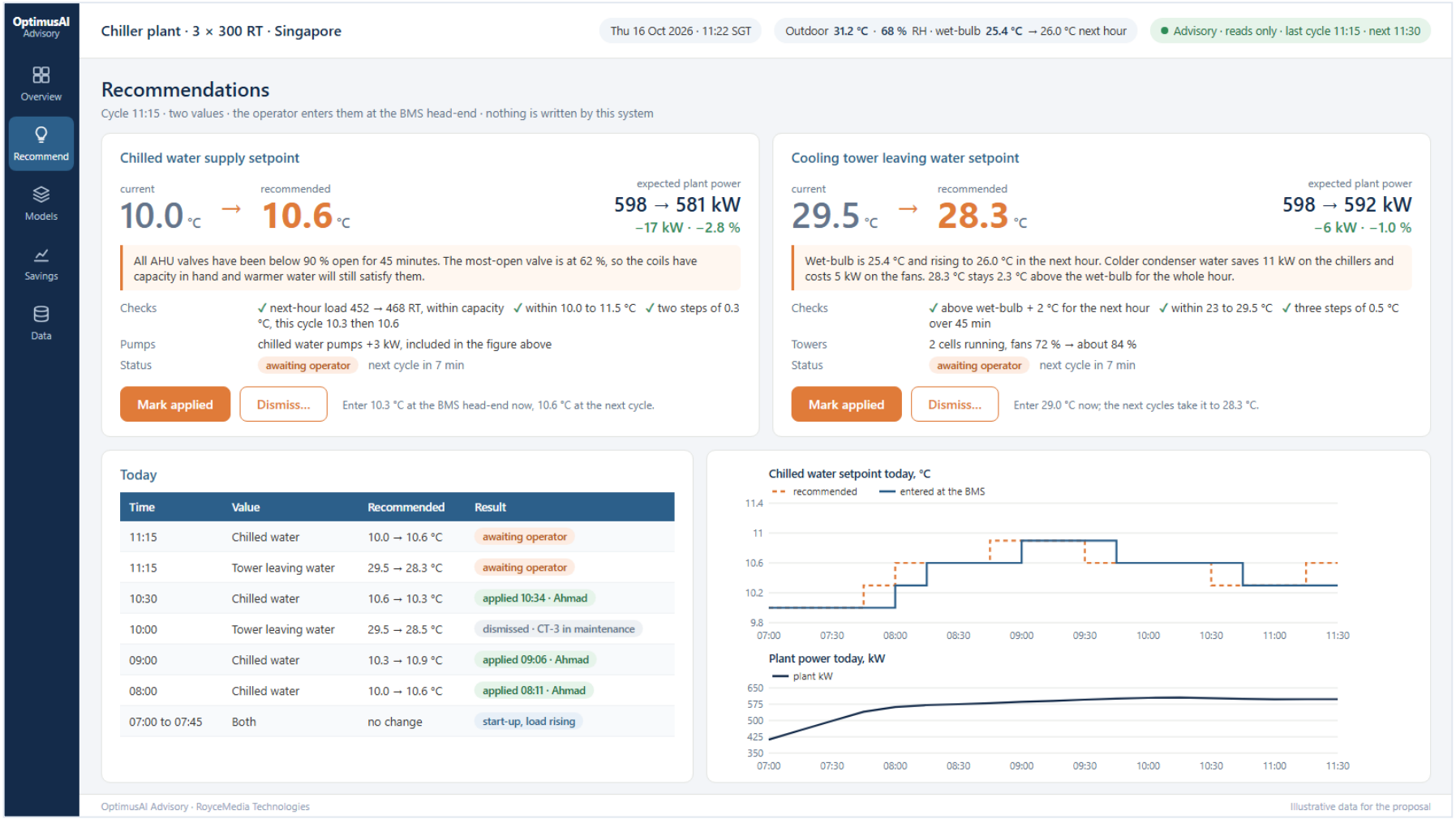
Chiller, tower, pump and fan curves give the kW of every candidate setpoint.

4 Recommend

The pair with the lowest kW, inside the limits, goes to the operator screen.

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What the operator sees: two values, two buttons



1 Current and recommended

One value per setpoint, with the expected plant kW before and after. The pumps and fans are already in the figure.

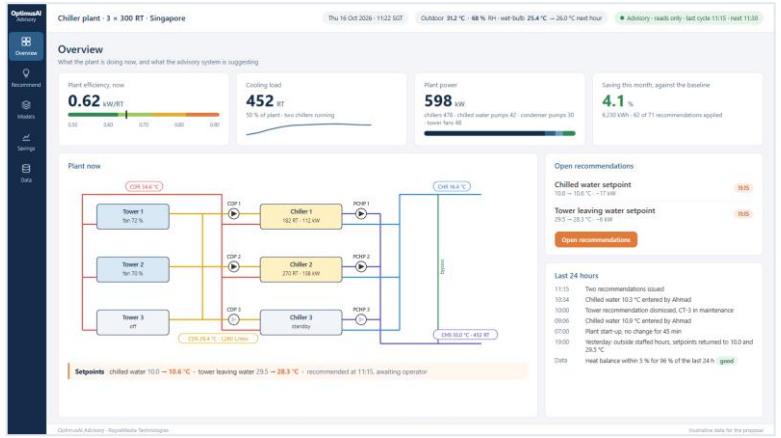
2 Why, with the checks

The signal behind it, in plain words: valve positions, wet-bulb, the next hour's load. Each limit shown as passed.

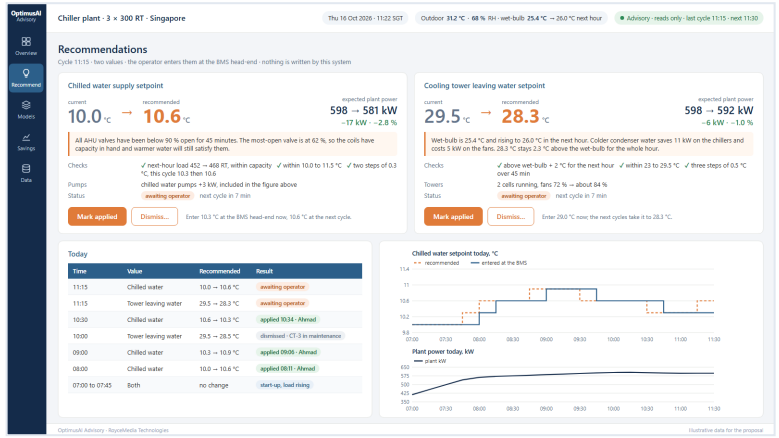
3 Mark applied or Dismiss

The operator enters the value at the BMS head-end and marks it applied. A dismissal is logged with its reason.

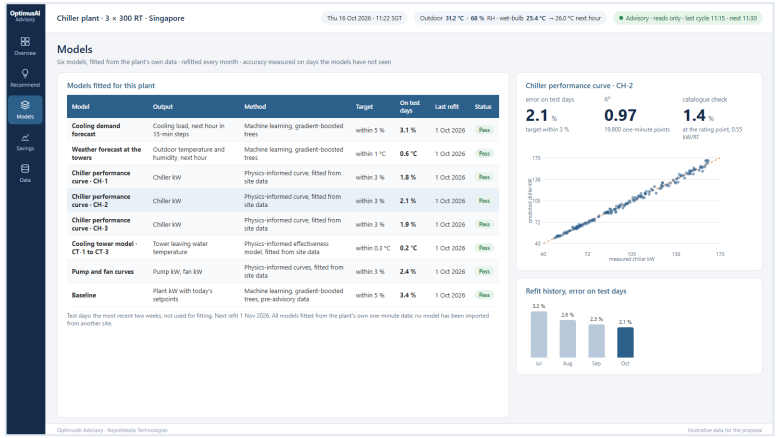
Five screens, one job: help the operator choose well



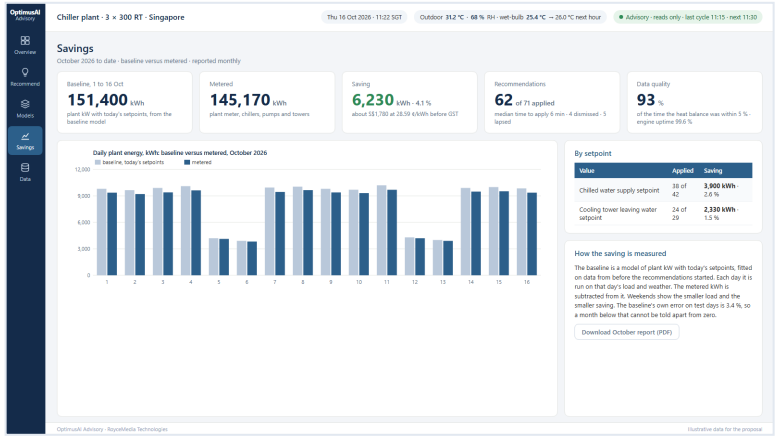
Overview: plant now, open recommendations



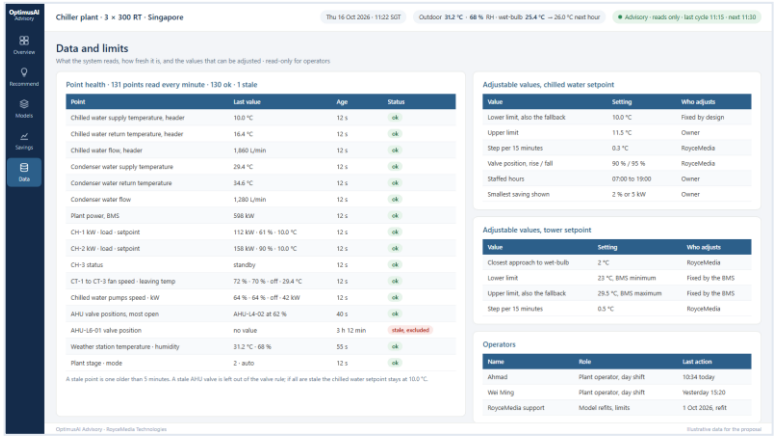
Recommendations: the two values



Models: what was fitted, how accurate



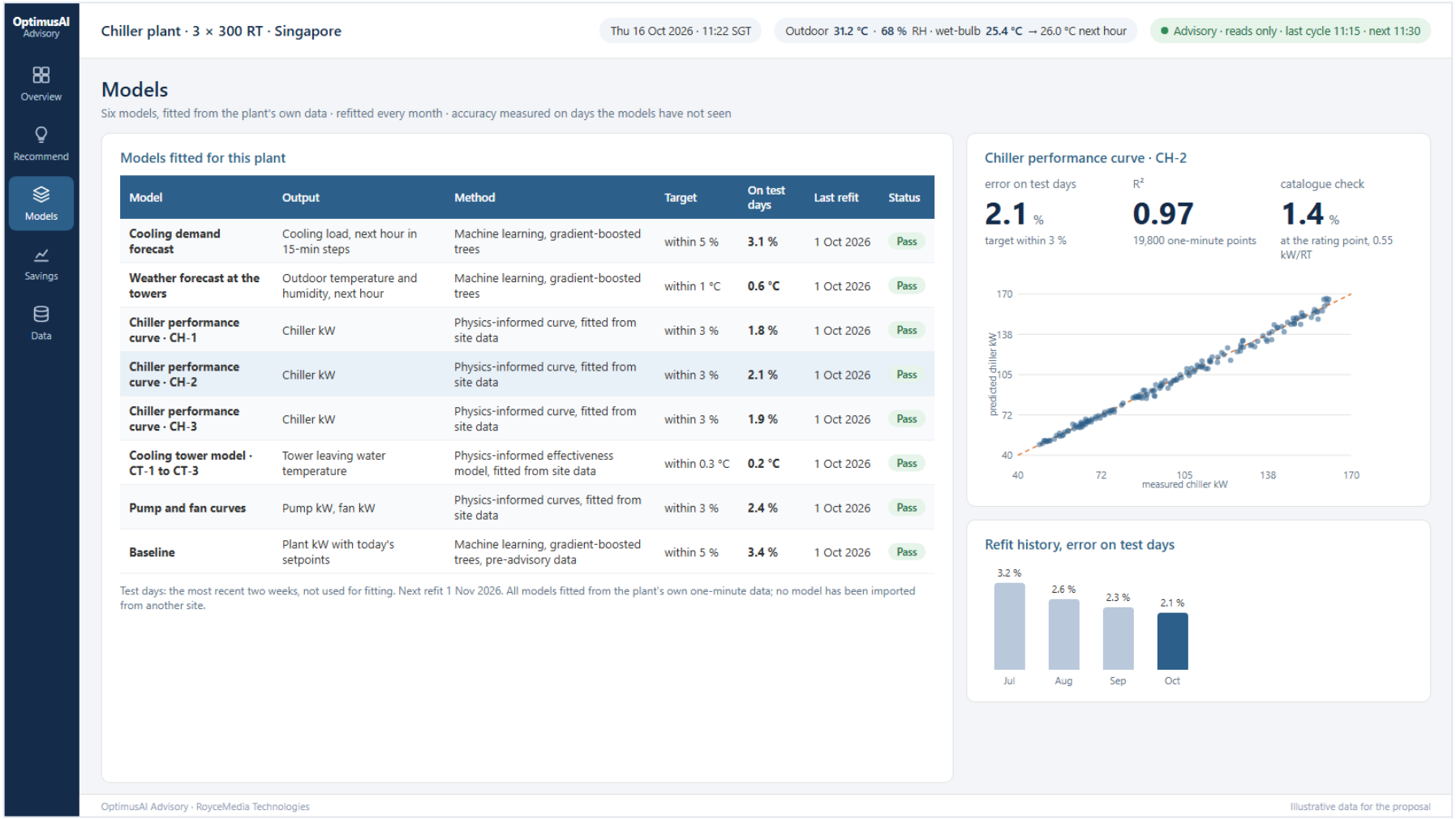
Savings: baseline versus metered, monthly



Data and limits: point health, adjustable values

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Six models, fitted from the plant's own data



1 Forecasts

Cooling load and weather for the next hour: machine learning on the plant's history and the local weather station. No internet feed.

2 Plant curves

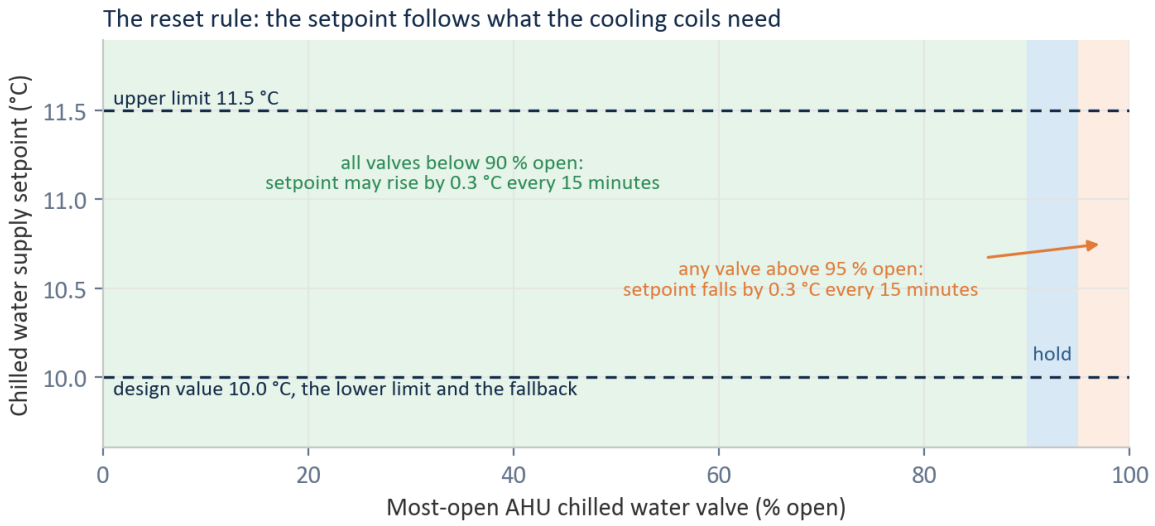
Chiller, tower, pump and fan: physics-informed curves fitted from site data, so a setpoint the plant has never run at can still be priced.

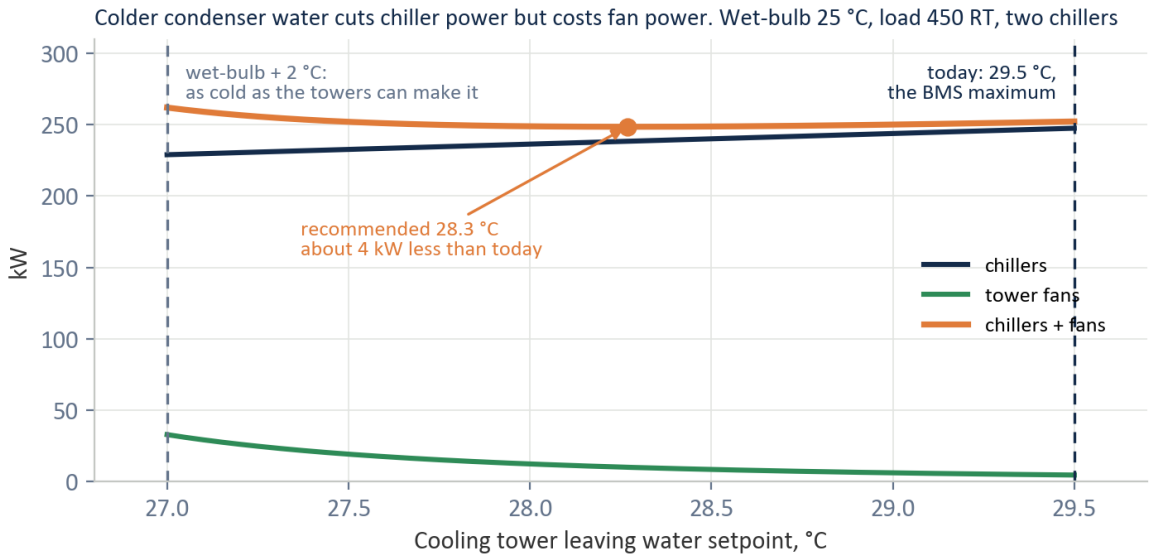
3 Baseline

Plant kW with today's setpoints, fitted on pre-advisory data. Every month's saving is measured against it.

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The rules: follow the coils, balance the chillers against the fans



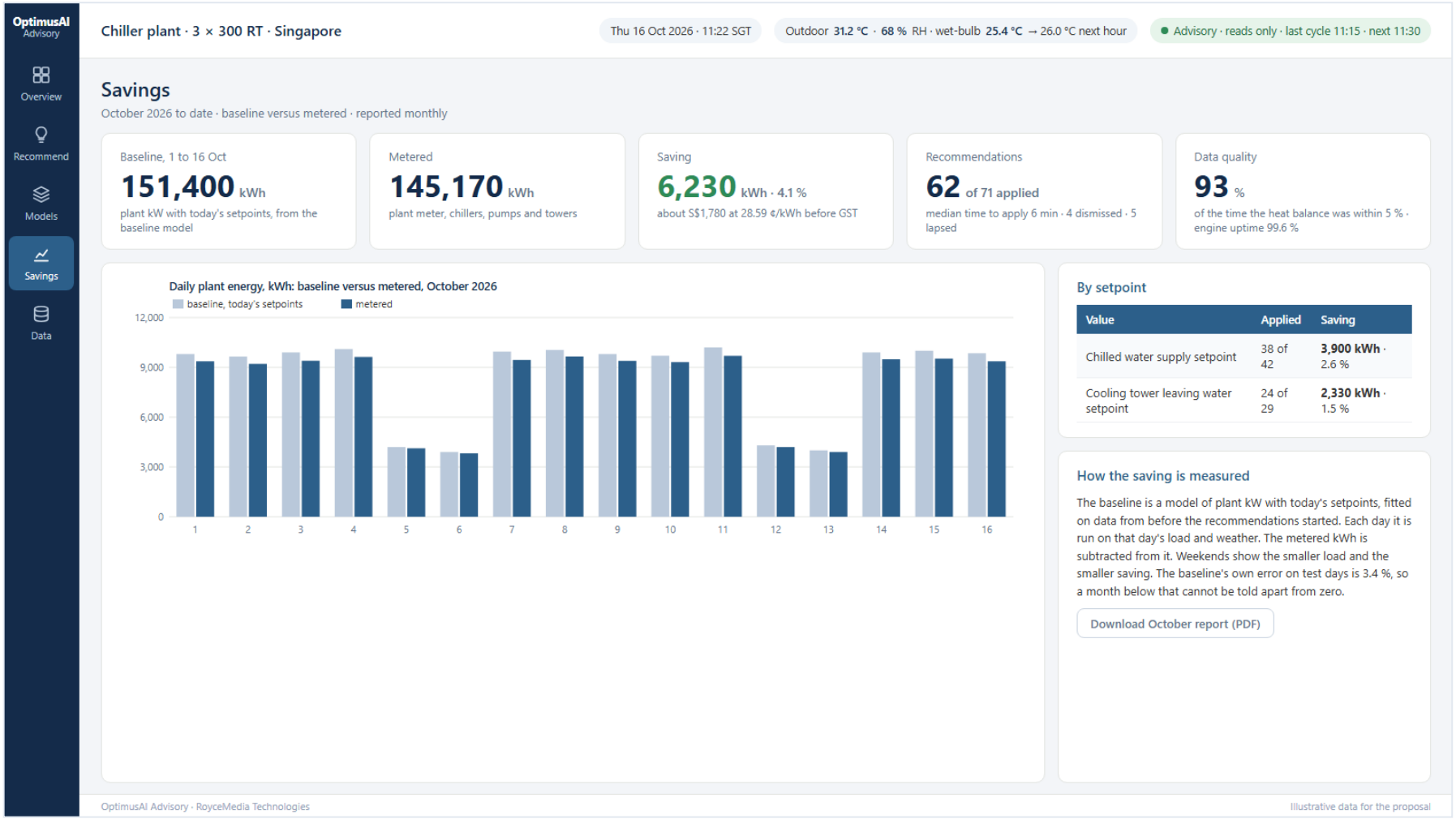


- All AHU valves below 90 % open for 15 minutes: the setpoint may rise 0.3 °C.
- Any valve above 95 %: it falls 0.3 °C. Between: hold.
- Never above 11.5 °C, never below the design value of 10.0 °C.

- Try each tower setpoint between wet-bulb + 2 °C and 29.5 °C, in 0.5 °C steps.
- Keep the one with the lowest chiller plus fan power.
- Never below 23 °C or above 29.5 °C, the BMS limits. At most 0.5 °C per 15 minutes.

Safeguards. Chiller fault or outside staffed hours: today's values. Stale or disagreeing sensors: no recommendation. Setpoint in manual at the BMS: no recommendation for it. A dismissed recommendation is logged and not repeated until conditions change.

Proving the saving: baseline versus the plant meter



1 The baseline

A model of plant kW with today's setpoints, fitted on data from before the recommendations started.

2 Each month

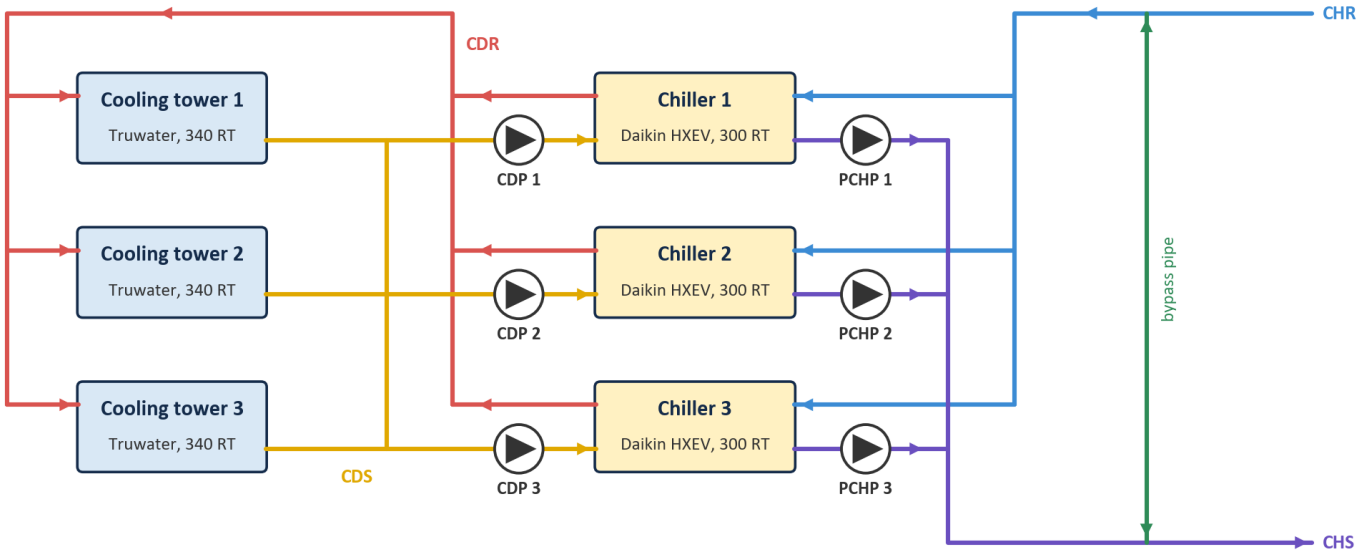
The baseline is run on that month's load and weather. The metered kWh is subtracted from it. The difference is the saving.

3 Honest limits

The baseline's own error is reported. A month whose saving is smaller than that error is reported as zero.

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What the site needs: existing points, two confirmations



Read once a minute, read-only

- Header temperatures and flows, both loops
- Each chiller: status, kW, load, water temperatures, setpoint
- Pumps and tower cells: status, speed, kW
- AHU chilled water valve positions
- Weather station at the towers
- Plant stage, mode, manual flags

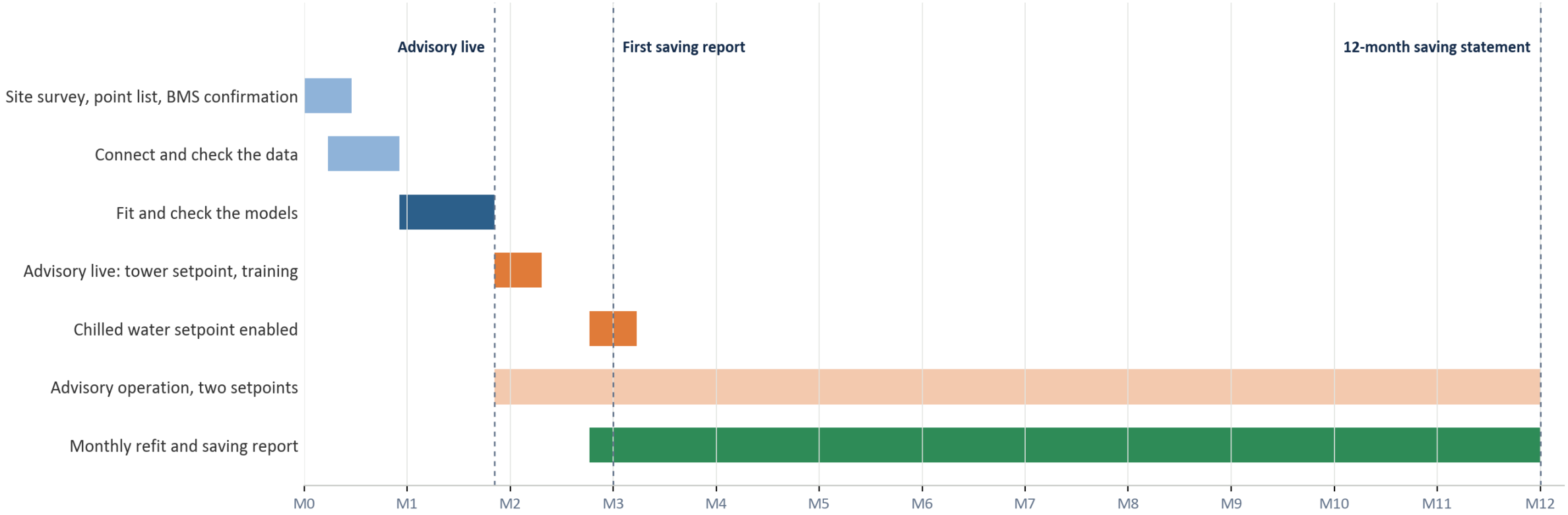
Data Assumptions

- Both setpoints can be entered at the head-end
- AHU valve positions are on the same BMS
- The BMS point names, from the point list request
- The logged data to date, for the first fit

One server on the plant network. No internet, no cloud, no new instruments. The drawing shows a three-chiller plant; the system fits any count.

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Advisory live in about two months, first saving report in three



Months from kick-off. The chilled water setpoint is enabled once the AHU valve positions are confirmed on the BMS; the tower setpoint needs no new points.

1 **Survey and point list** weeks 1 to 2

2 **Connect and check the data** weeks 2 to 4

3 **Fit and check the models** weeks 5 to 8

4 **Advisory live, tower setpoint** week 8

5 **Chilled water setpoint enabled** week 12

6 **Monthly refit and saving report** from week 12

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Assumptions

Sensors are read as they are. No calibration by us. A sensor that is off puts the recommendations off by the same amount.

The weather station is the only weather source. No internet. If it reads wrong, the wet-bulb and the forecasts are wrong with it.

The models know only what the plant has done. The first fit uses the data logged so far; accuracy improves with each monthly refit.

The valve positions are trusted. A valve reading 100 % holds the chilled water at 10 °C, the safe side. A valve reading low could allow a rise the coil does not want.

The BMS stays as described. The limits here follow the control description. If the BMS contractor changes a sequence or threshold, the limits are updated with it.

The saving depends on the operator. Nothing changes until the value is entered. The report counts what was entered.

Percentages are typical, not promised. The plant's own figures come from its data and are reported against the baseline every month.

The plant stays the same. A chiller replacement, a new tenant or new hours changes the baseline, and it is refitted from that point.

We deliver

- OptimusAI Software License
- The five operator screens
- Models fitted from the plant's data, refitted monthly
- The monthly saving report
- Operator training

The BMS keeps doing

- Staging, pump and fan control, alarms, interlocks
- Holding the setpoints the operator enters
- Everything in the control description, unchanged

Q&A

Thank you!

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